

# AI Design Generation Test Report — “Cobras” Theme

Date: June 2025

Tester: Automated QA via Besu Customs 3D Configurator

AI Provider: Google Gemini 3 Pro (2K Resolution)

Environment: Next.js Dev Server (localhost:3000)

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## Test Overview

All 28 3D models in the Besu Customs configurator were tested with the AI Design Generation feature using a consistent prompt to evaluate texture application quality, color accuracy, and model compatibility.

### Prompt Used

“Create a futuristic-looking uniform for a team named Cobras, utilizing the color palette of grey, black, and orange”

### Color Palette

| Color  | Hex     |
|--------|---------|
| Grey   | Various |
| Black  | #000000 |
| Orange | Various |

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## Test Results

All 28 models were tested successfully. The AI generated and applied textures to each 3D model in real-time.

### Status Summary

| Status                                     | Count |
|--------------------------------------------|-------|
| <input checked="" type="checkbox"/> Passed | 28    |
| <input type="checkbox"/> Failed            | 0     |
| Total                                      | 28    |

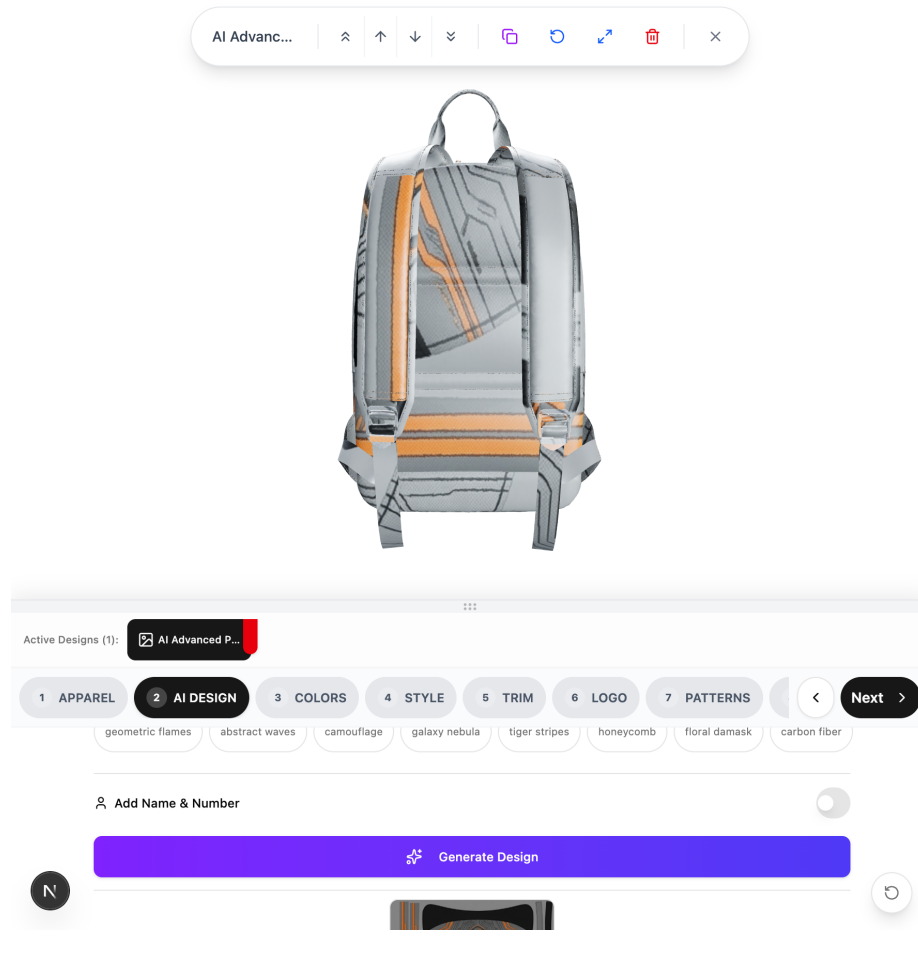
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## Individual Model Results

### 1. Backpack

Status: ☐ Generated Successfully

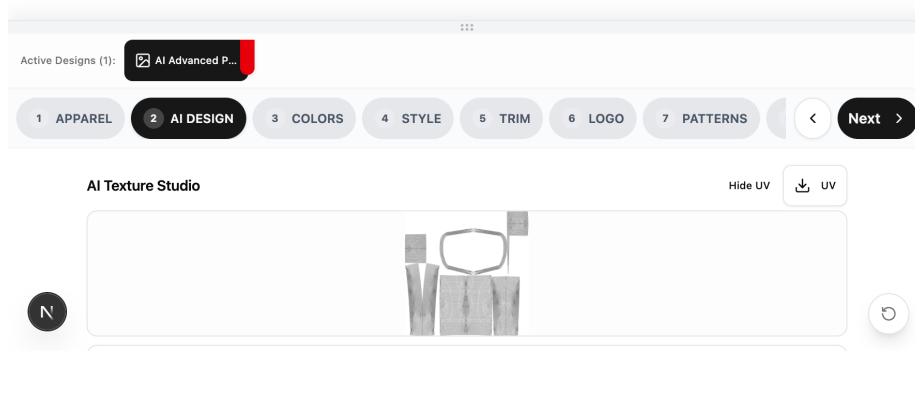
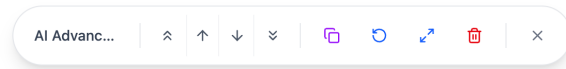
Category: Accessories



### 2. Duffle Bag

Status: ☐ Generated Successfully

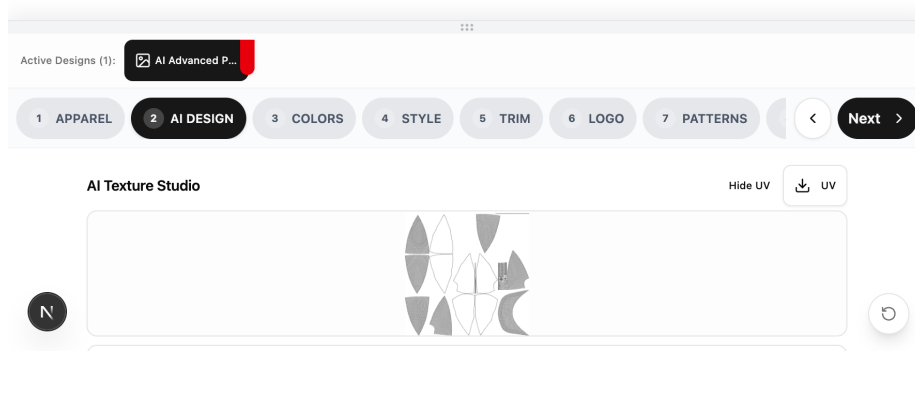
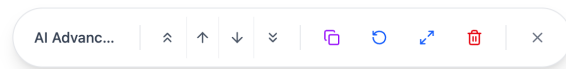
Category: Accessories



### 3. Baseball Caps

Status: ☐ Generated Successfully

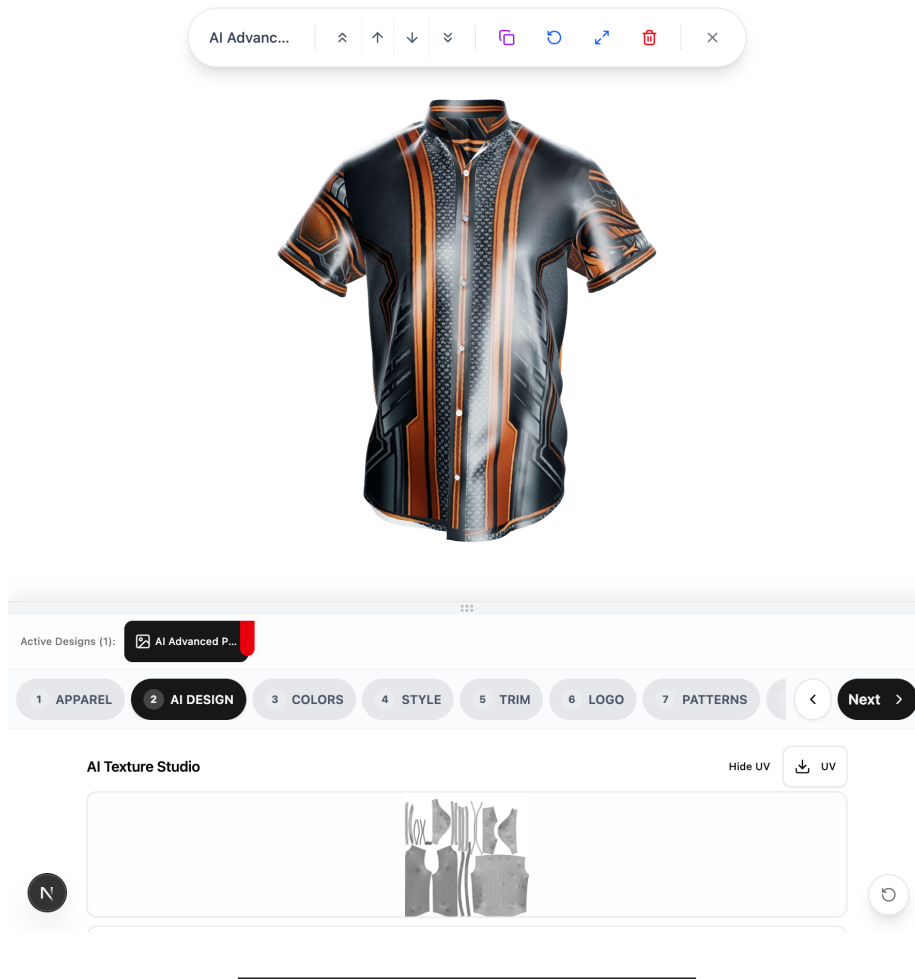
Category: Headwear



#### 4. Baseball Jersey

Status: ☐ Generated Successfully

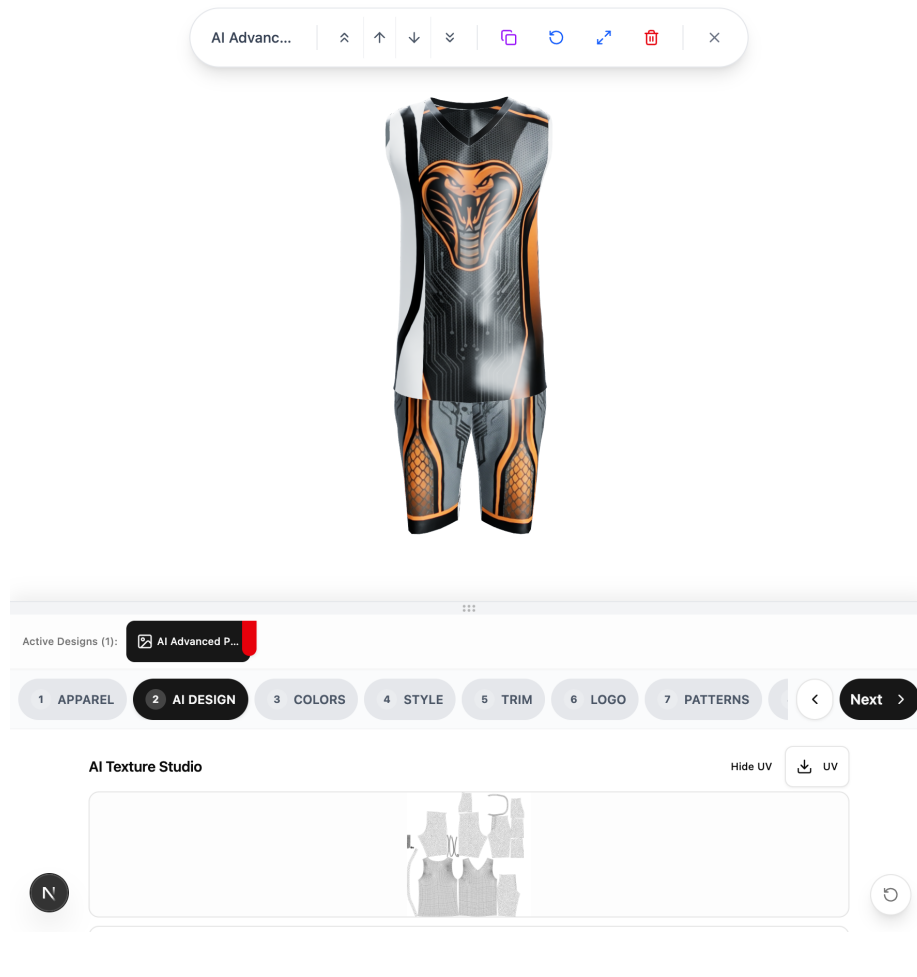
Category: Baseball



## 5. Basketball Jersey and Shorts

Status: ☐ Generated Successfully

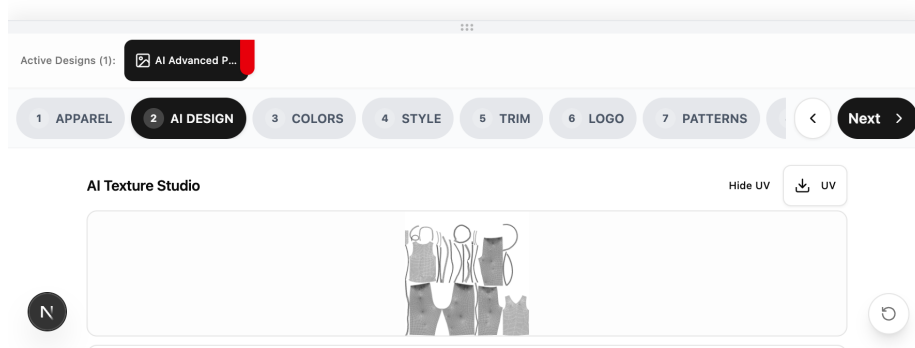
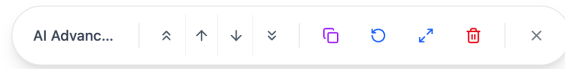
Category: Basketball



## 6. Basketball Jersey Top And Long Shorts

Status: ☐ Generated Successfully

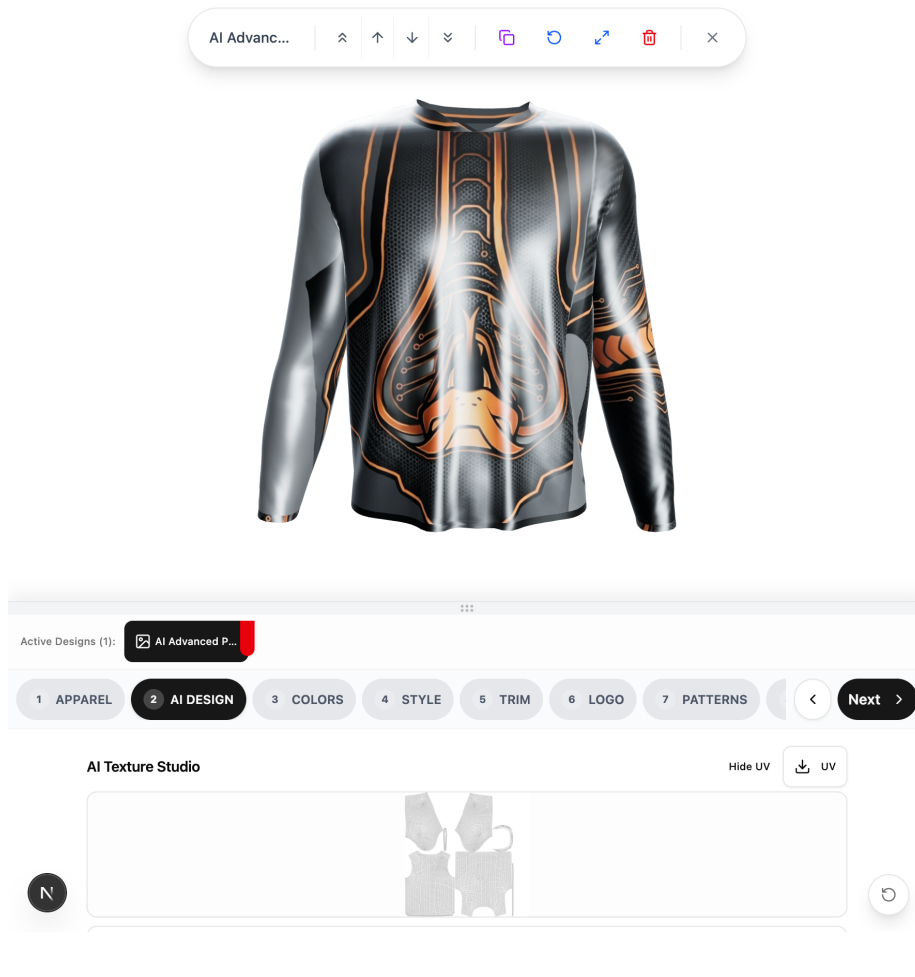
Category: Basketball



## 7. Basketball Shooting Shirt Long Sleeve

Status: ☐ Generated Successfully

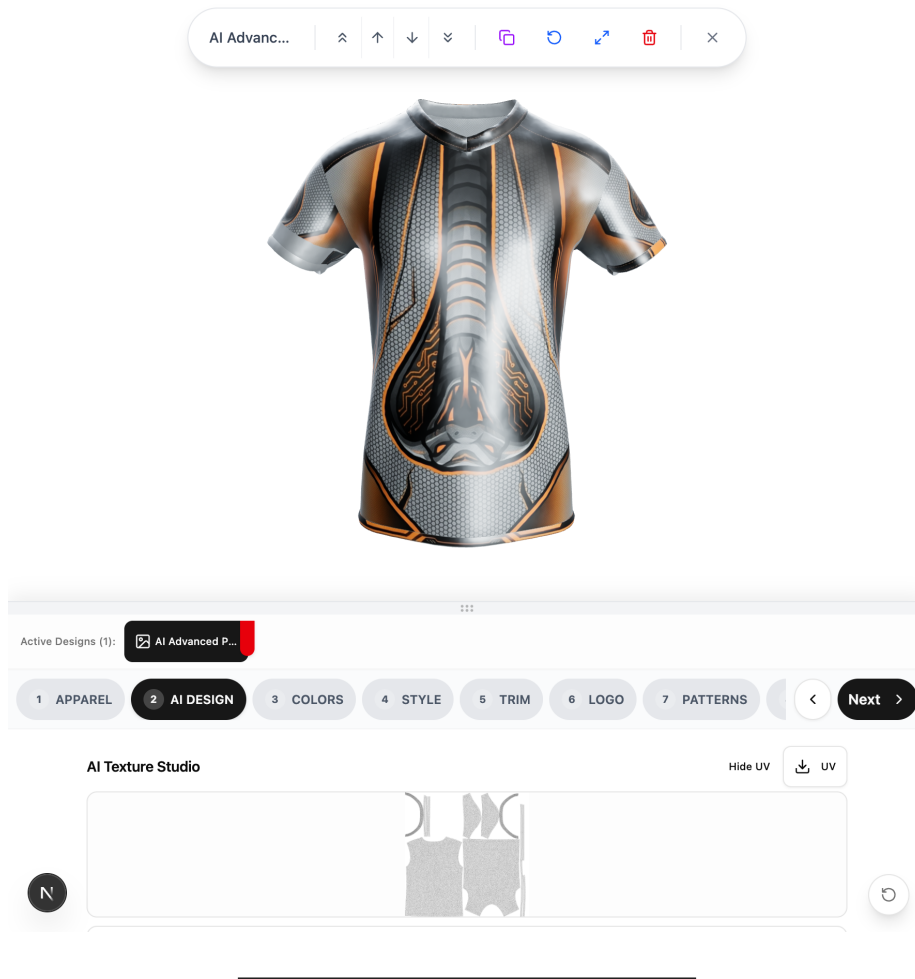
Category: Basketball



## 8. Basketball Shooting Shirt Short Sleeve

Status: ☐ Generated Successfully

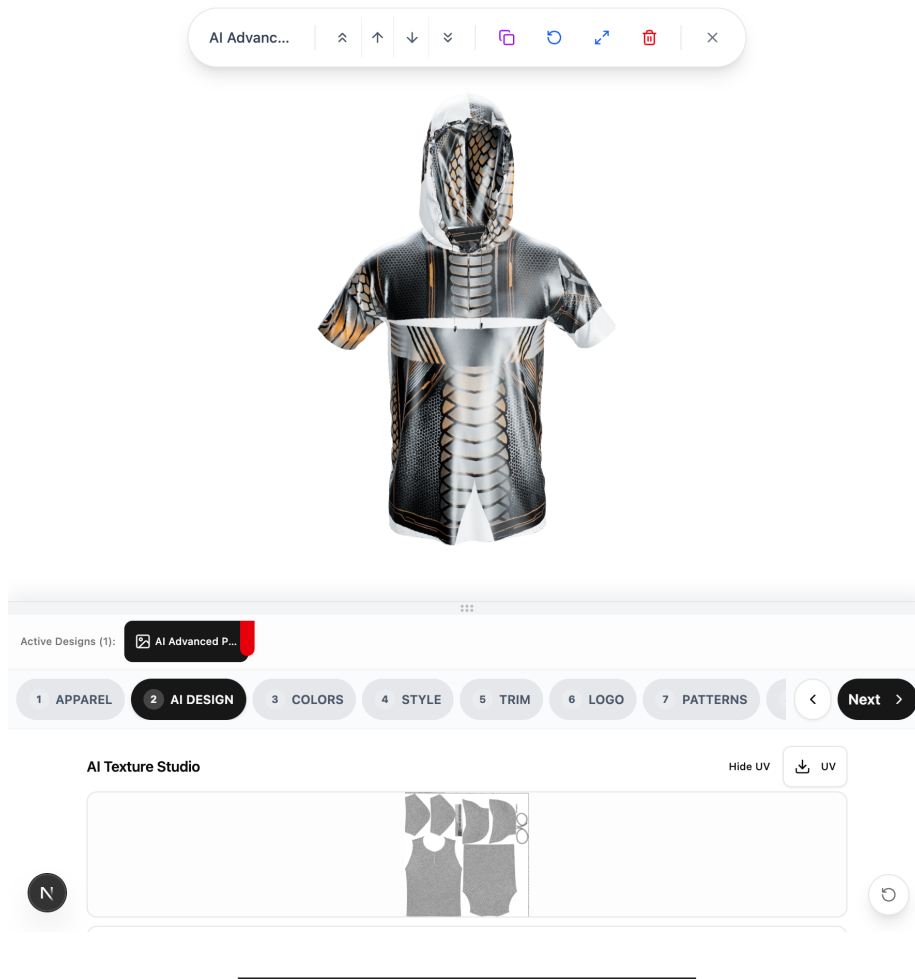
Category: Basketball



## 9. Basketball Shooting Shirt with Hoodie

Status: ☐ Generated Successfully

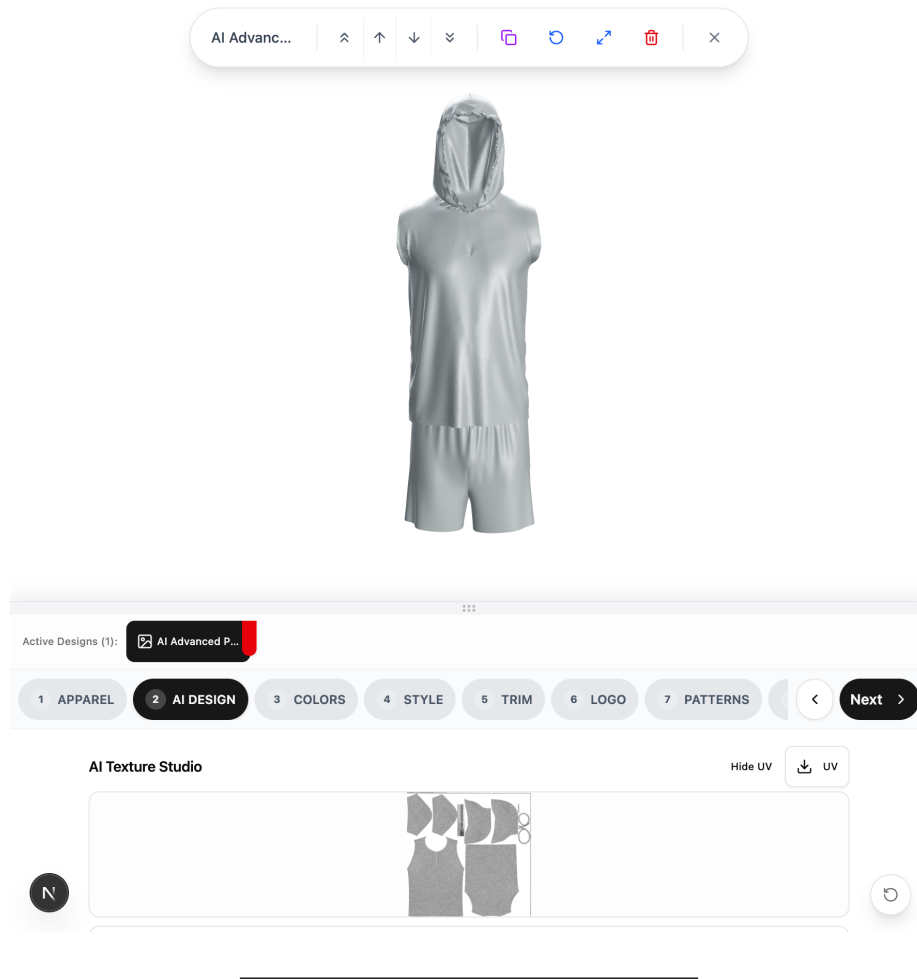
Category: Basketball



## 10. Flag Football Jersey with Hoodie

Status: ☐ Generated Successfully

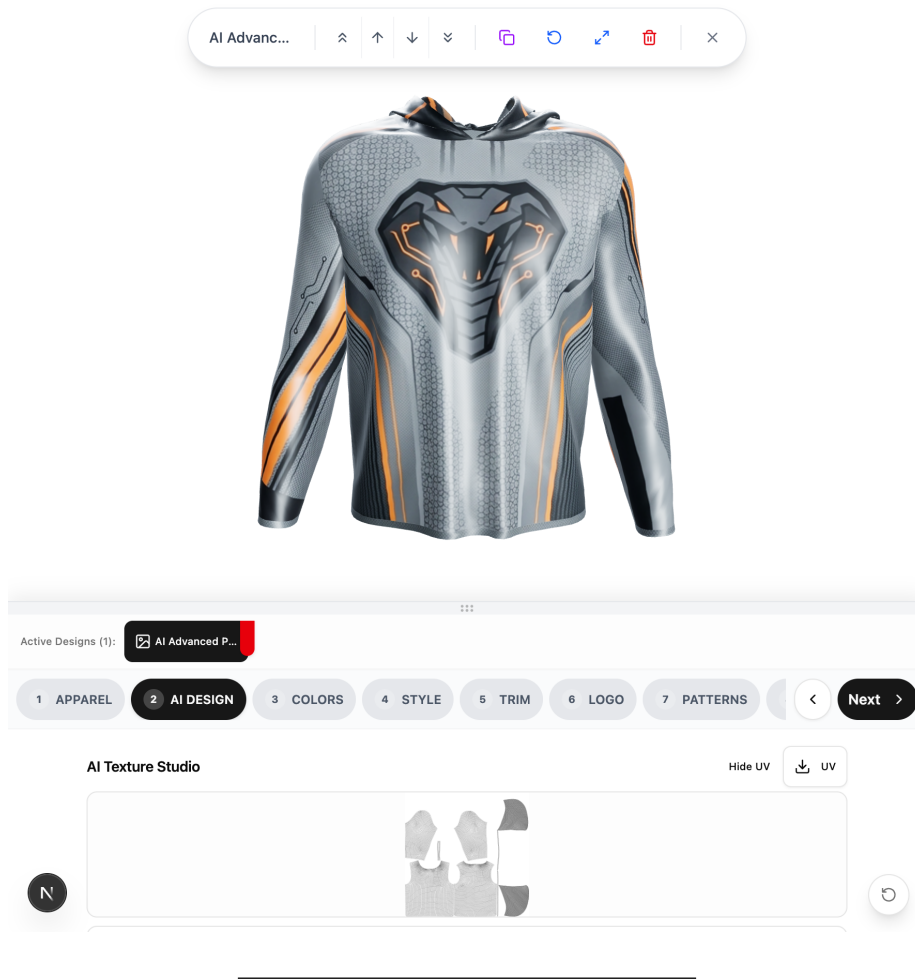
Category: Football



## 11. Hoodie

Status: ☐ Generated Successfully

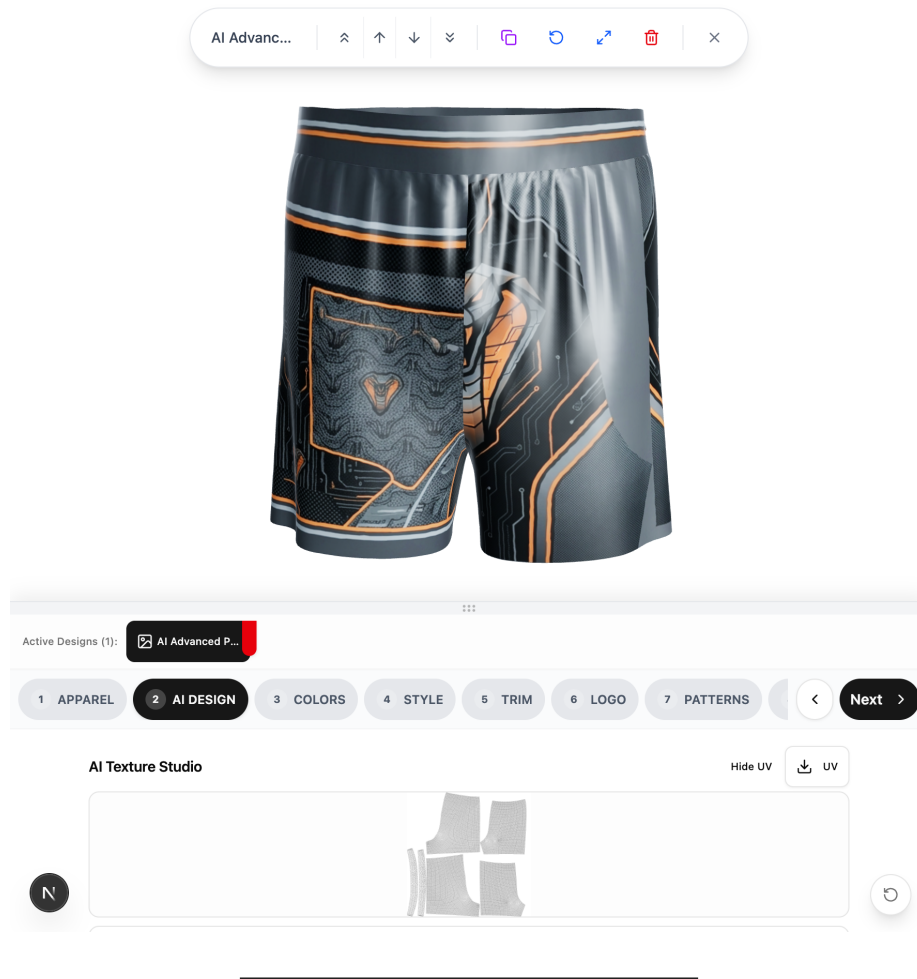
Category: General Apparel



## 12. Half Size Shorts

Status: ☐ Generated Successfully

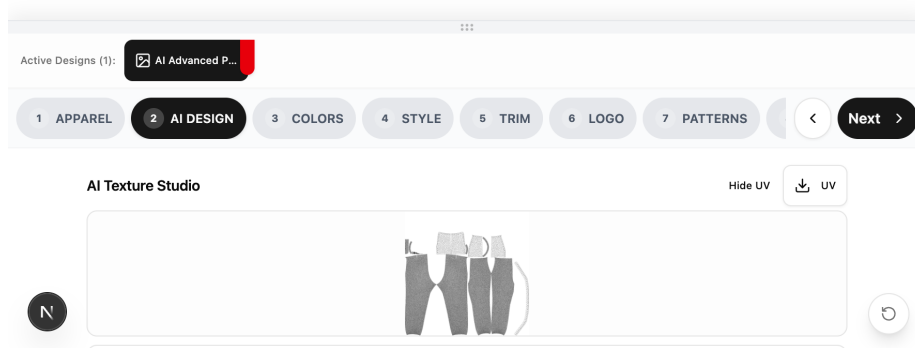
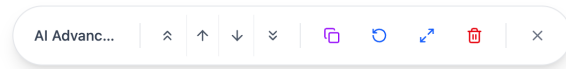
Category: Bottoms



### 13. Standard Bottom Cut, Cuffed

Status: ☐ Generated Successfully

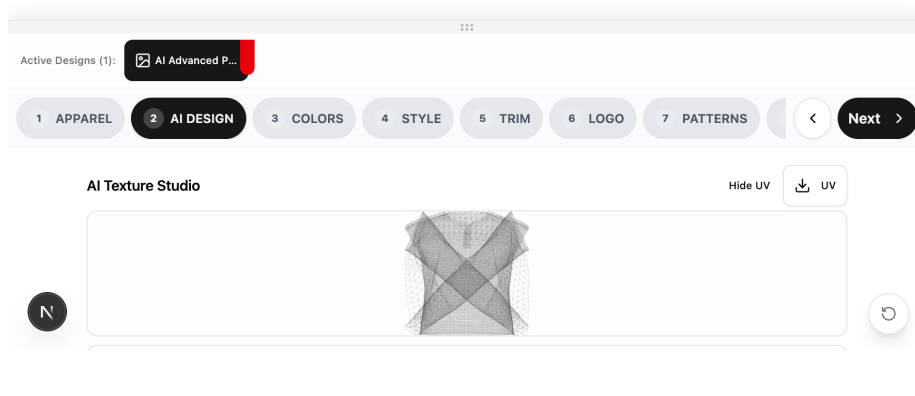
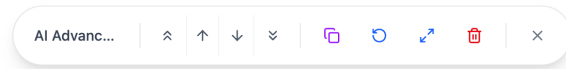
Category: Bottoms



#### 14. Polo Shirts Long Sleeve

Status: ☐ Generated Successfully

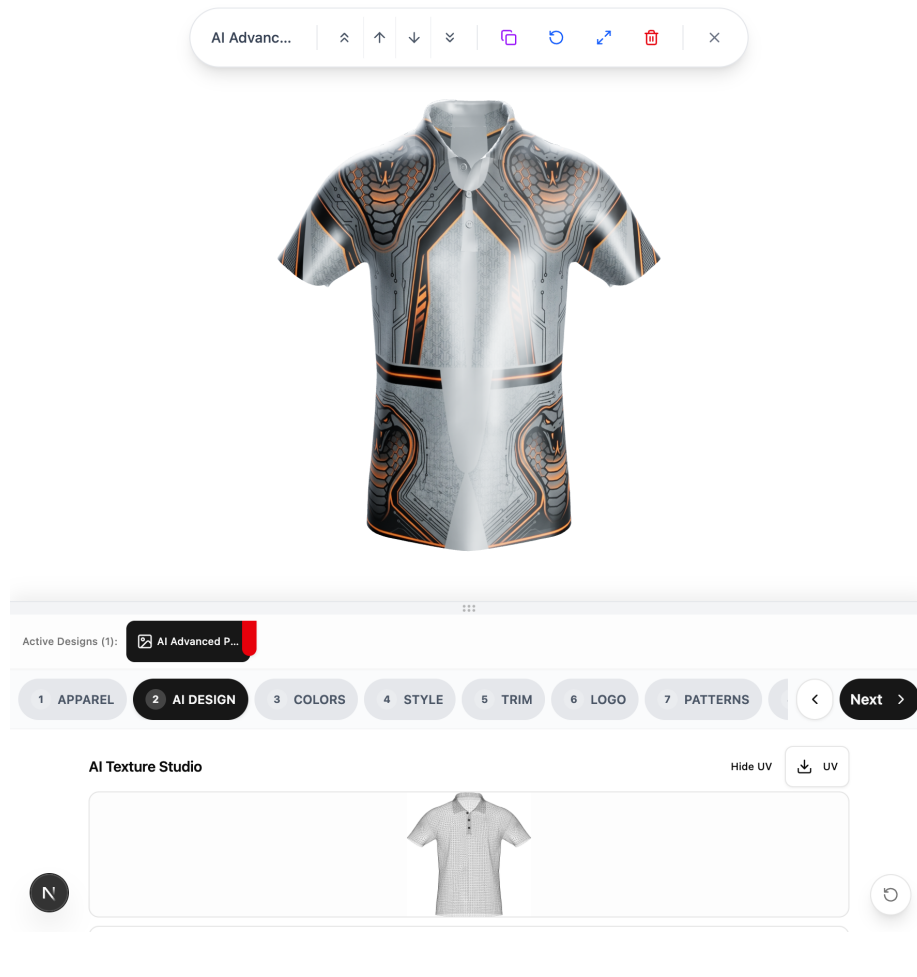
Category: Polo



## 15. Polo Shirts Short Sleeve

Status: ☐ Generated Successfully

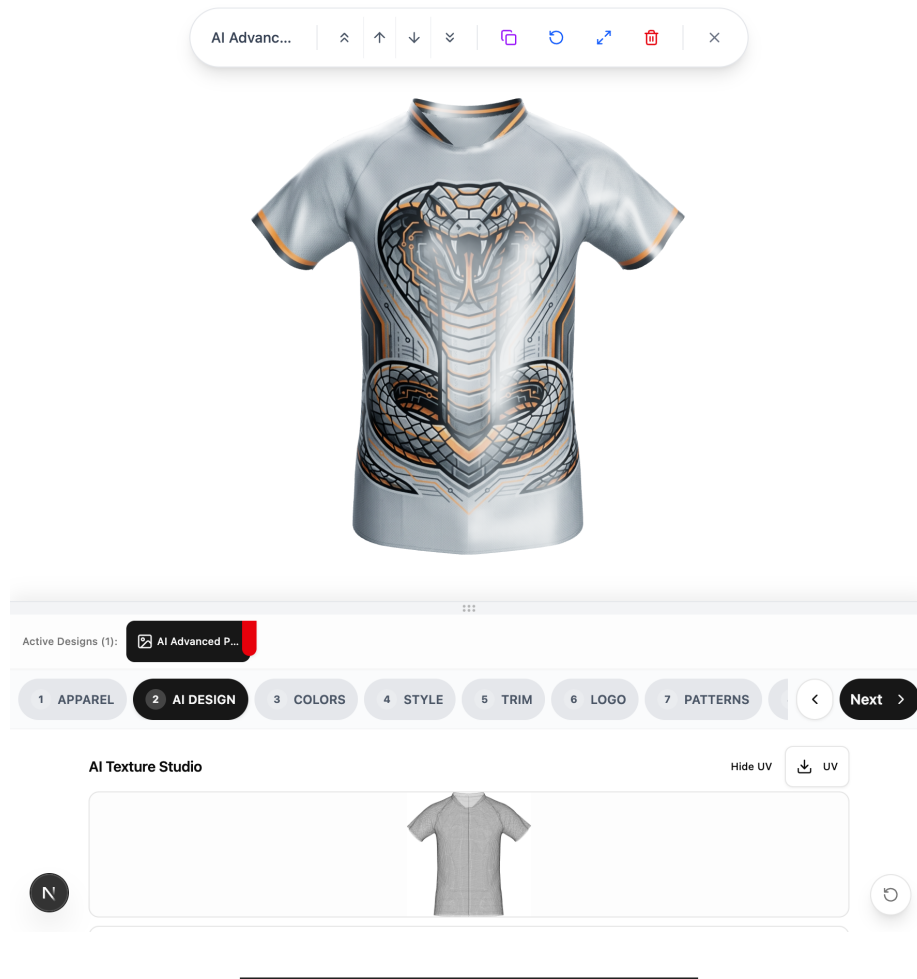
Category: Polo



## 16. Soccer Jersey Crew Neck

Status: ☐ Generated Successfully

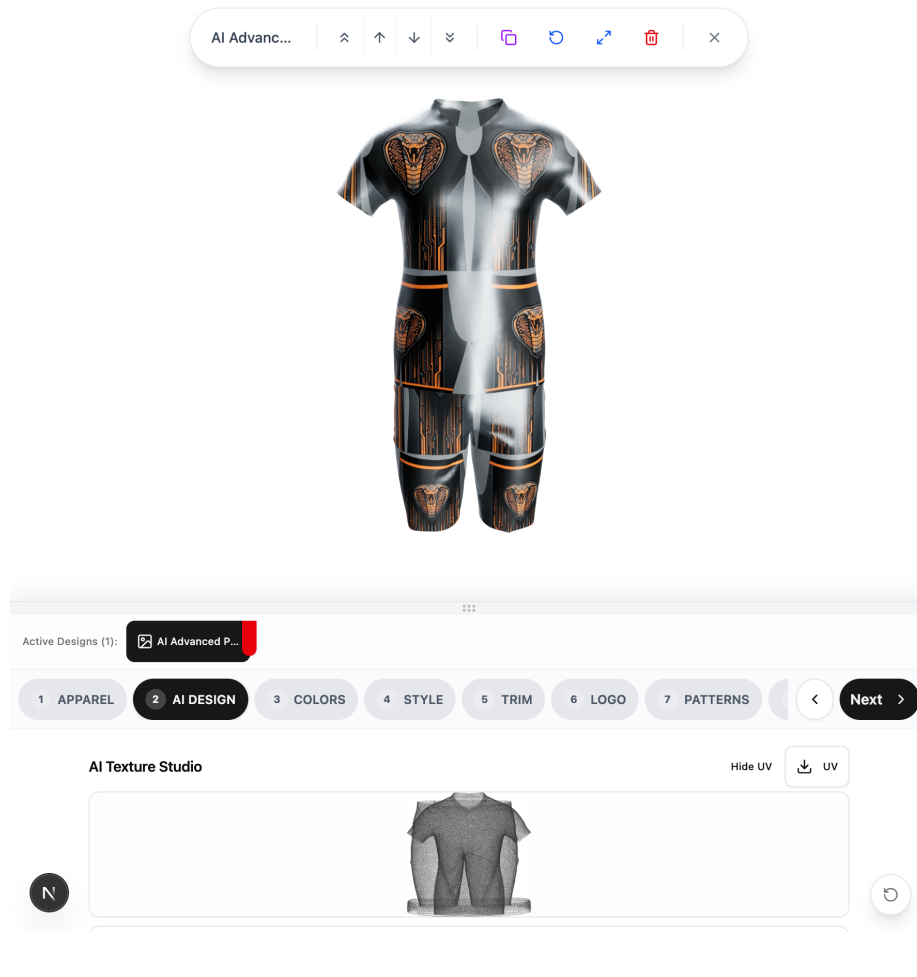
Category: Soccer



### 17. Soccer Jersey V-Neck

Status: ☐ Generated Successfully

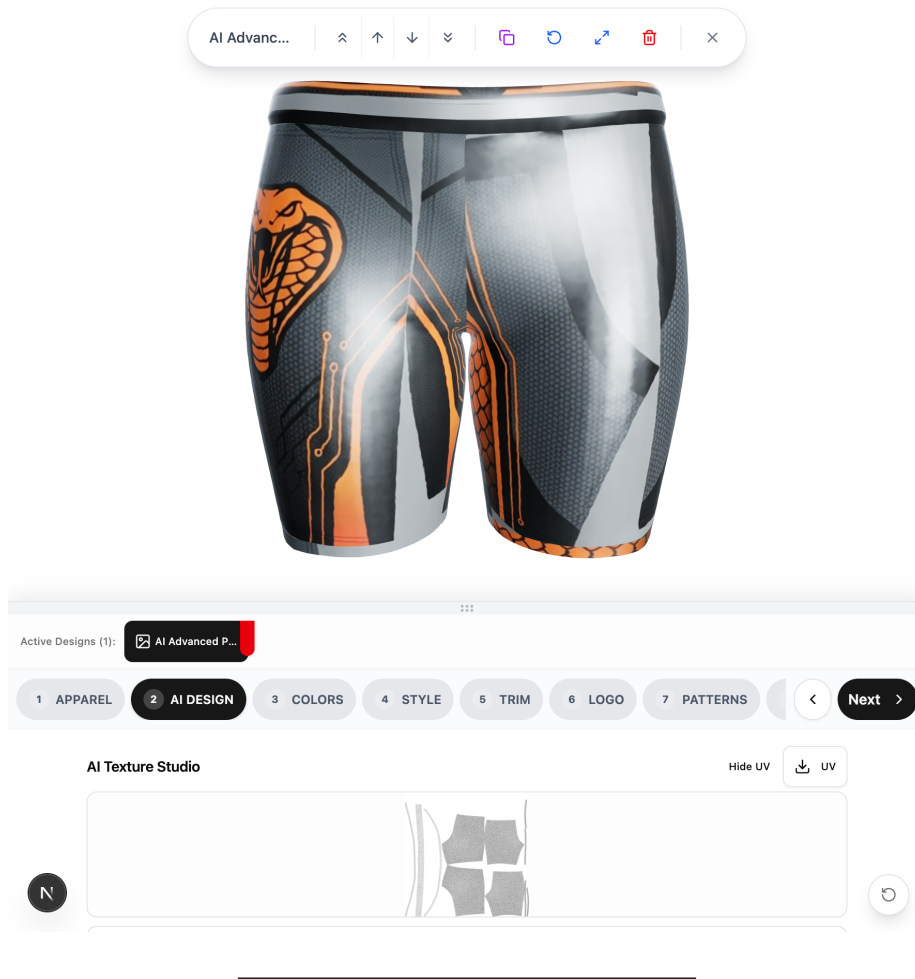
Category: Soccer



## 18. Track & Field Compression Shorts

Status: ☐ Generated Successfully

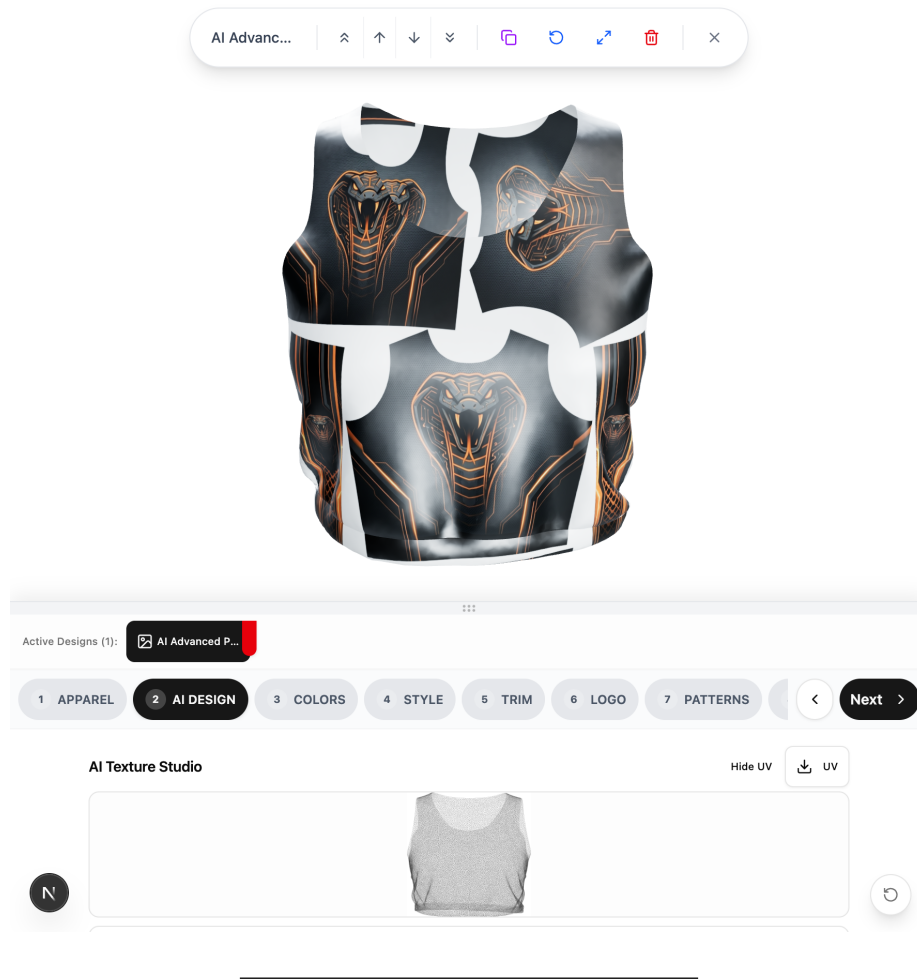
Category: Track & Field



## 19. Track & Field Crop Top

Status: ☐ Generated Successfully

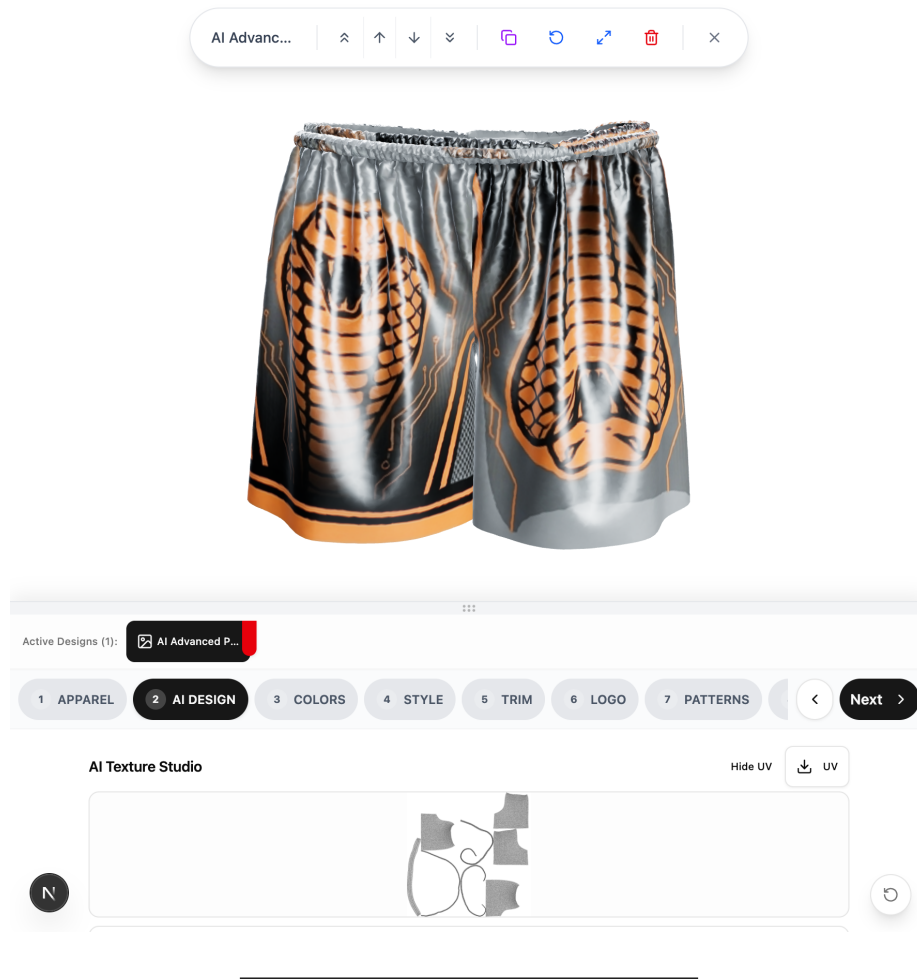
Category: Track & Field



## 20. Track & Field Mid-Length Shorts

Status: ☐ Generated Successfully

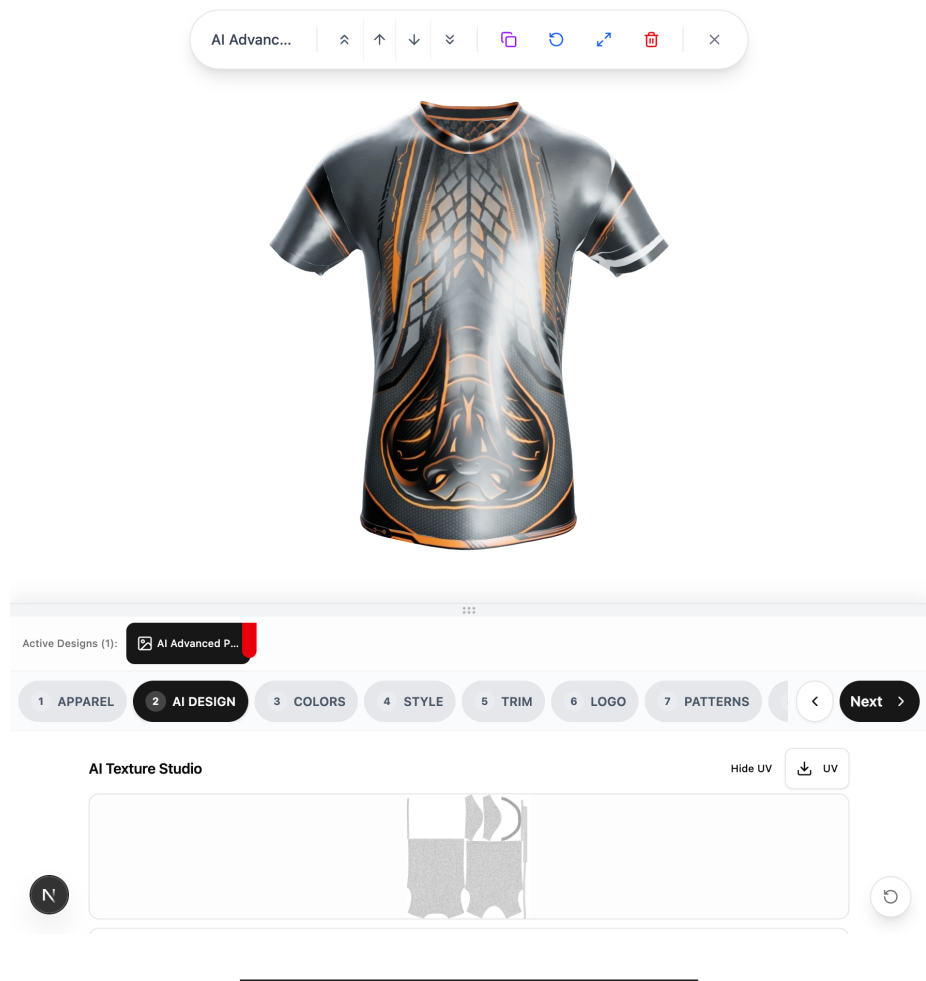
Category: Track & Field



## 21. Track & Field Short Sleeve

Status: ☐ Generated Successfully

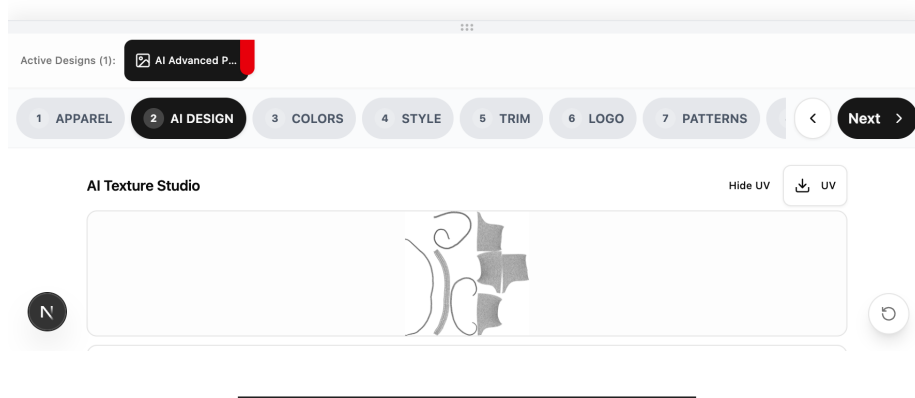
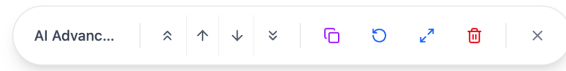
Category: Track & Field



## 22. Track & Field Split Shorts

Status: ☐ Generated Successfully

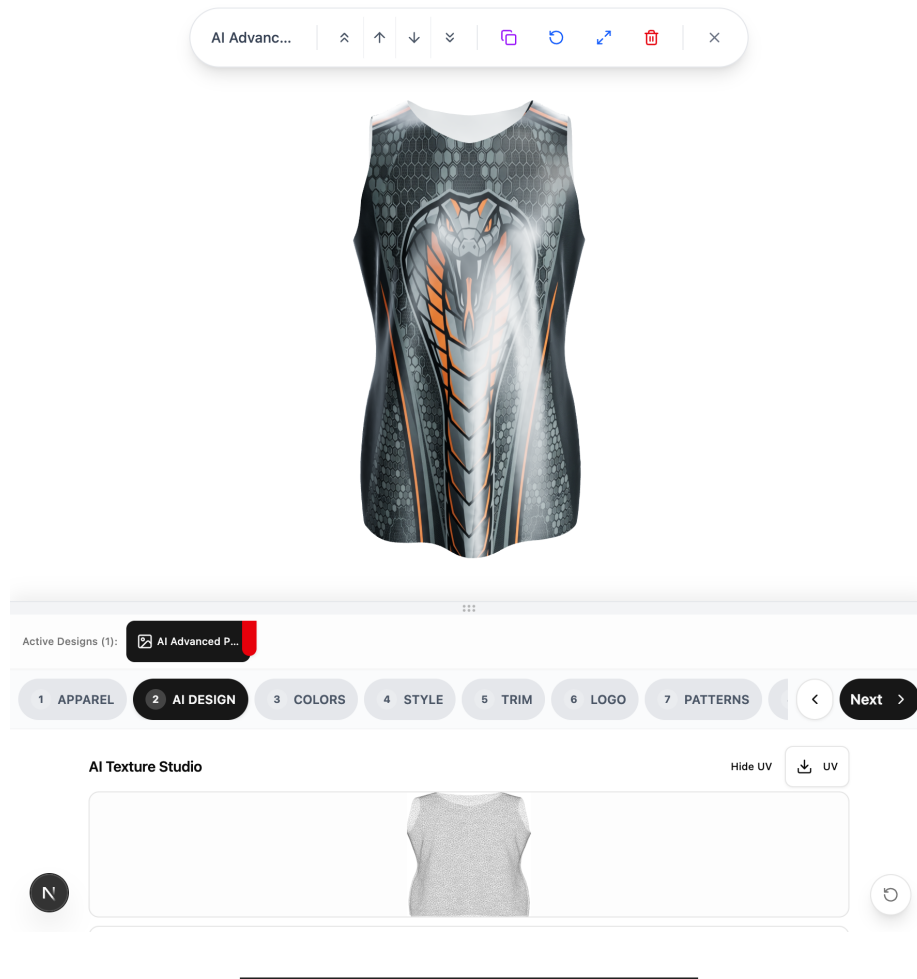
Category: Track & Field



## 23. Track & Field Tank Top

Status: ☐ Generated Successfully

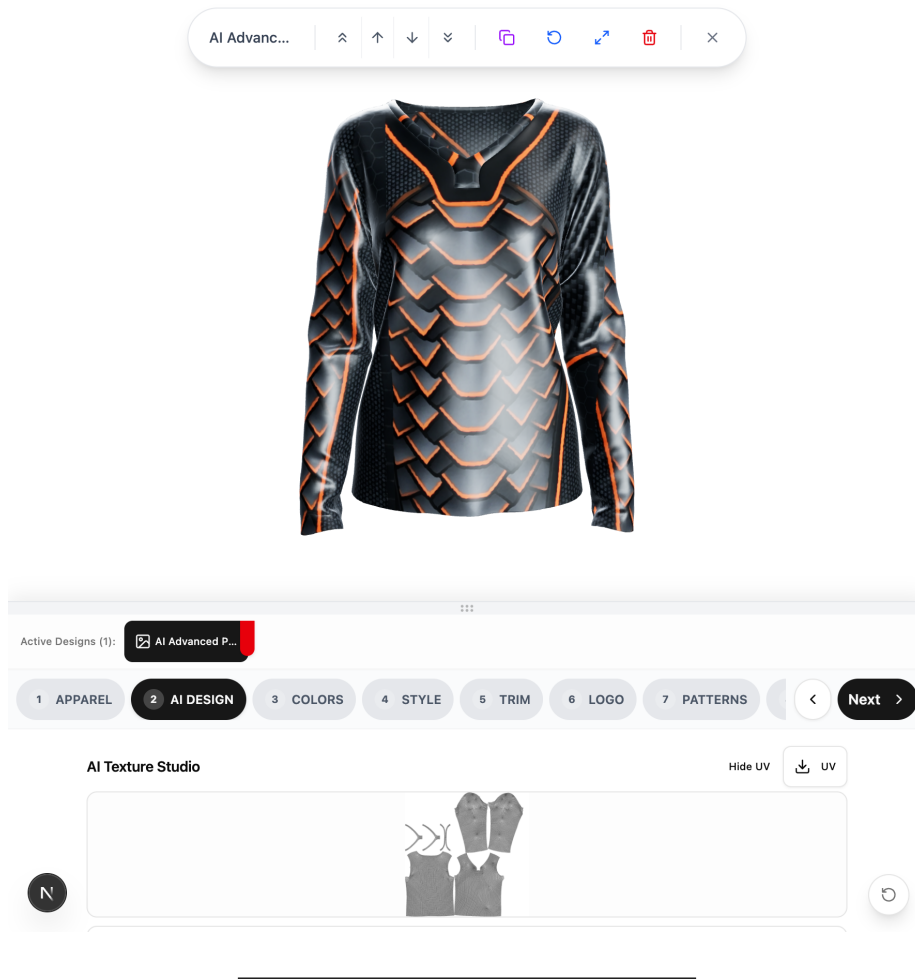
Category: Track & Field



## 24. Volleyball Long Sleeve Tops

Status: ☐ Generated Successfully

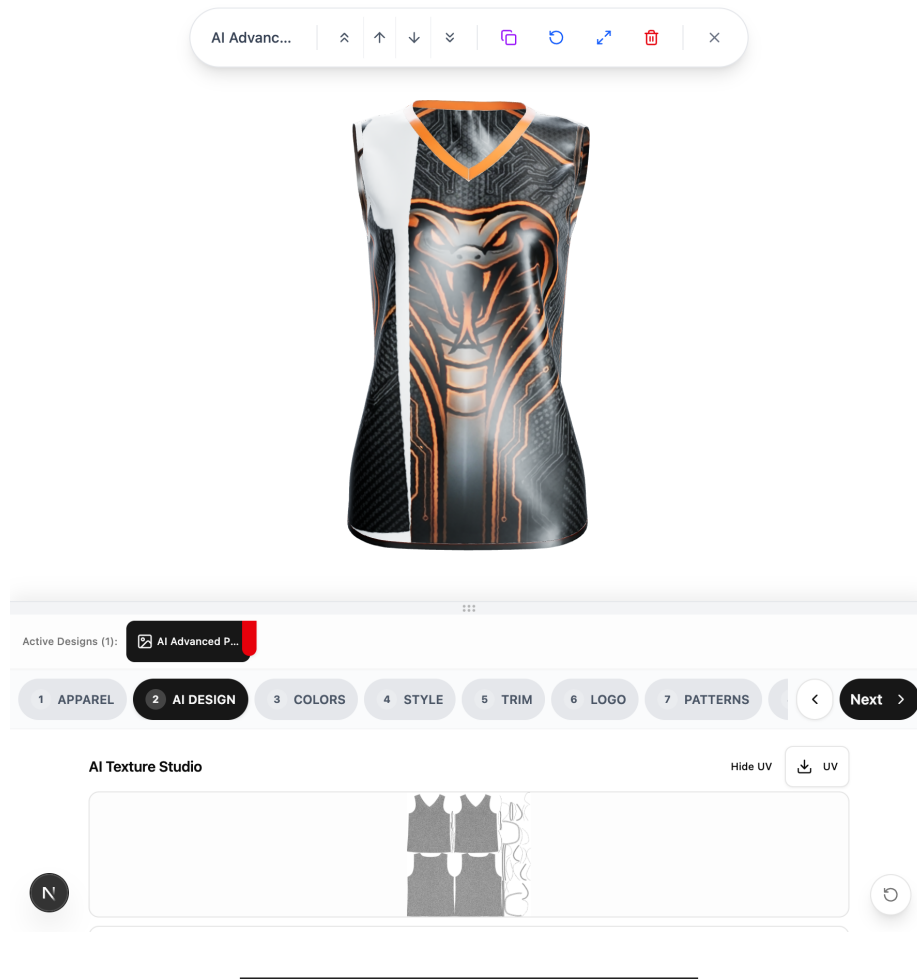
Category: Volleyball



## 25. Volleyball Short Sleeve Tops

Status: ☐ Generated Successfully

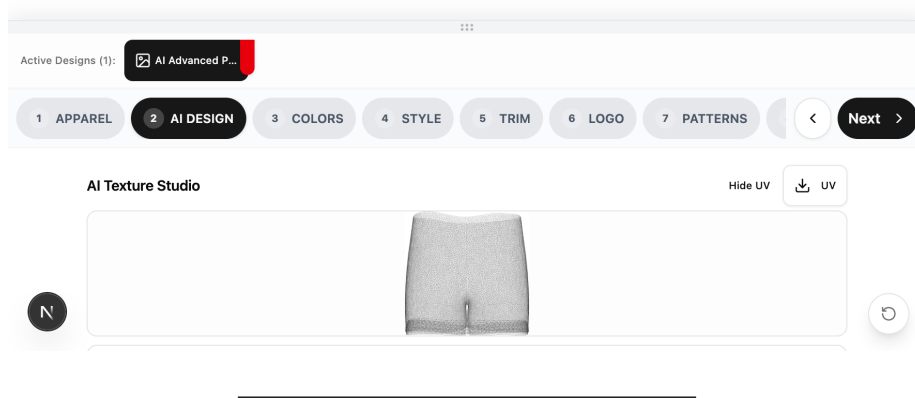
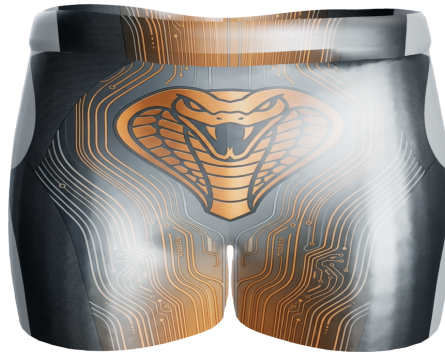
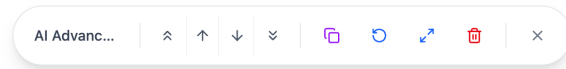
Category: Volleyball



## 26. Volleyball Shorts Spandex

Status: ☐ Generated Successfully

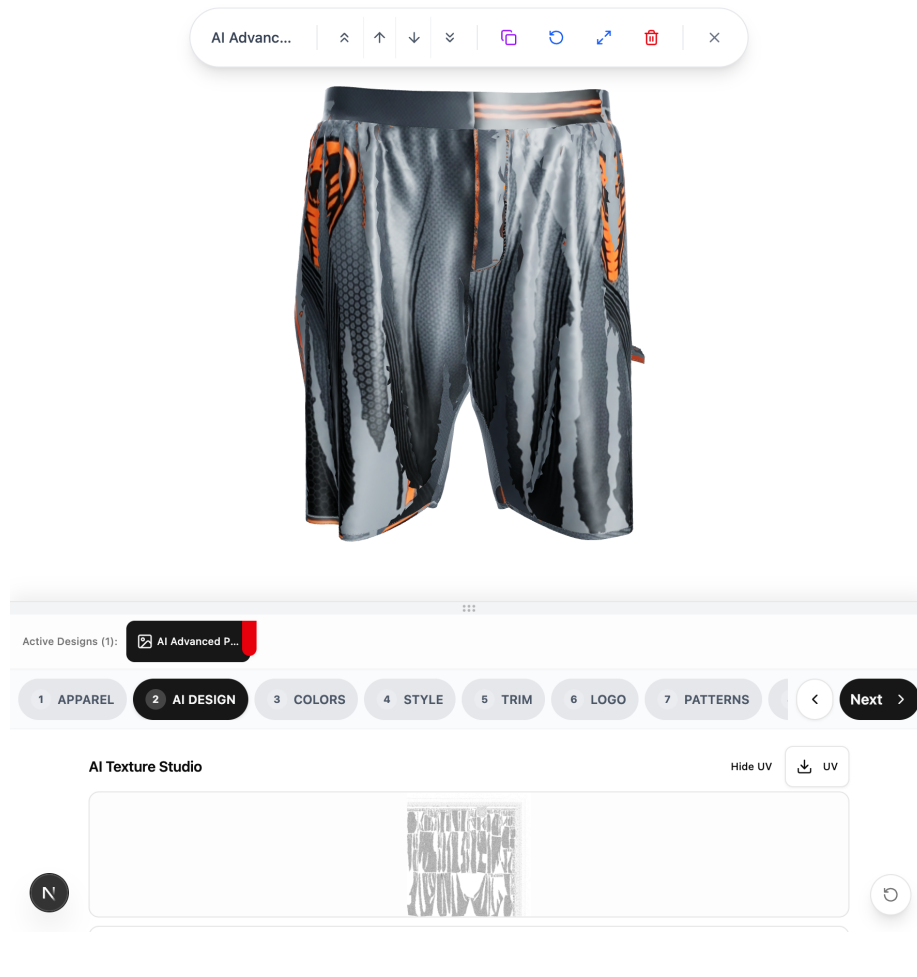
Category: Volleyball



## 27. Volleyball Shorts Spandex 4

Status: ☐ Generated Successfully

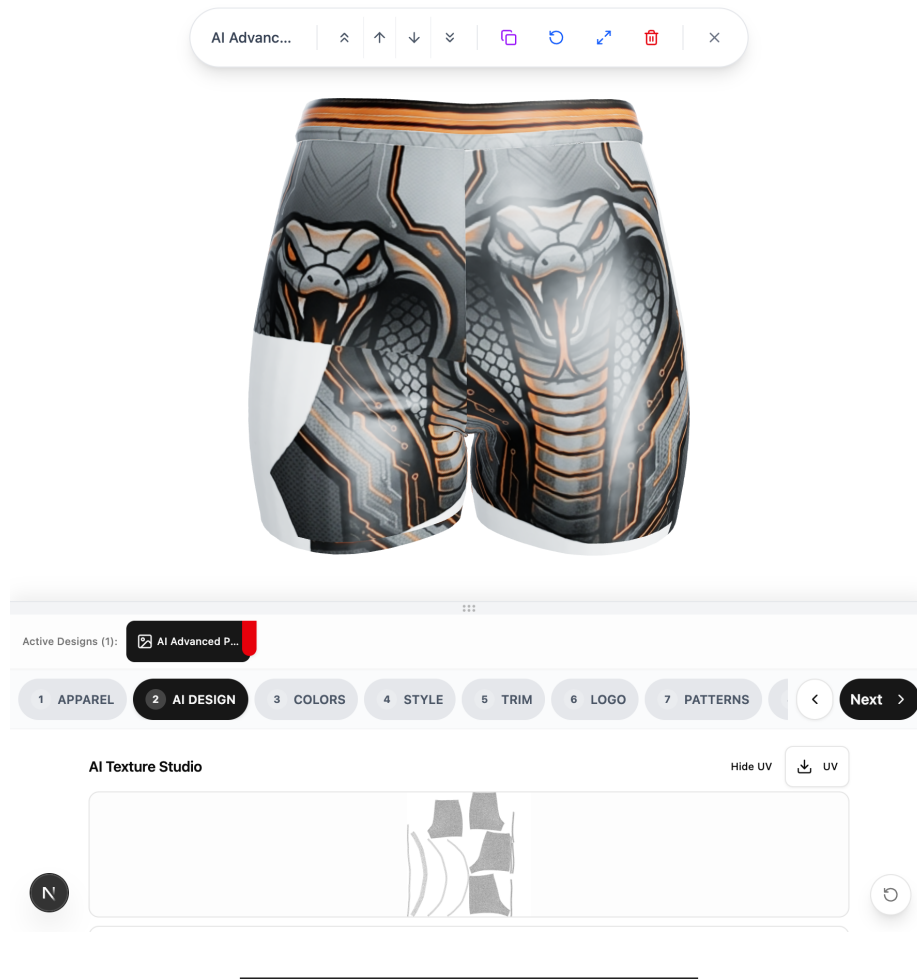
Category: Volleyball



## 28. Volleyball Spandex

Status: ☐ Generated Successfully

Category: Volleyball



## Technical Details

### AI Generation Pipeline

1. Input: Text prompt describing desired design
2. AI Provider: Google Gemini 3 Pro generates a 2K texture image
3. Application: Texture is UV-mapped onto the 3D model in real-time via Three.js
4. Rendering: React Three Fiber renders the textured model with PBR materials

### Bug Fix Applied (Prior to Testing)

A race condition was identified and fixed in the texture application pipeline:  
- File: lib/three-material-utils.ts — Added hasActiveTextureLayers guard to prevent applyMaterialsToThreeModel() from clearing the texture map when the TextureCompositor is managing textures - File: components/ai-texture-generator.tsx — Removed dead useEffect that redundantly called applyGoogle(scene) - Result: Textures now persist correctly on all 28 models

### Technology Stack

| Component          | Technology                   |
|--------------------|------------------------------|
| Frontend           | Next.js 16                   |
| 3D Engine          | Three.js + React Three Fiber |
| AI Provider        | Google Gemini 3 Pro          |
| State Management   | Zustand (persisted)          |
| Texture Resolution | 2048 x 2048 (2K)             |

### Categories Breakdown

| Category        | Models                                                                         | Count |
|-----------------|--------------------------------------------------------------------------------|-------|
| Accessories     | Backpack, Duffle Bag                                                           | 2     |
| Headwear        | Baseball Caps                                                                  | 1     |
| Baseball        | Baseball Jersey                                                                | 1     |
| Basketball      | Jersey+Shorts,<br>Jersey+Long Shorts,<br>Shooting Shirt<br>(Long/Short/Hoodie) | 5     |
| Football        | Flag Football Jersey<br>with Hoodie                                            | 1     |
| General Apparel | Hoodie                                                                         | 1     |
| Bottoms         | Half Size Shorts,<br>Standard Bottom Cut<br>Cuffed                             | 2     |
| Polo            | Long Sleeve, Short<br>Sleeve                                                   | 2     |
| Soccer          | Crew Neck, V-Neck                                                              | 2     |

| Category      | Models                                                                                | Count |
|---------------|---------------------------------------------------------------------------------------|-------|
| Track & Field | Compression Shorts, Crop Top, Mid-Length Shorts, Short Sleeve, Split Shorts, Tank Top | 6     |
| Volleyball    | Long Sleeve, Short Sleeve, Shorts Spandex, Shorts Spandex 4, Spandex                  | 5     |
| Total         |                                                                                       | 28    |

## Conclusion

The AI Design Generation feature is fully functional across all 28 3D models in the Besu Customs configurator. The “Cobras” futuristic design with grey, black, and orange color palette was successfully generated and applied to every model category — from accessories (backpacks, bags, caps) to full uniforms (basketball, soccer, volleyball, track & field) and individual apparel pieces (hoodies, polos, shorts).

Key Findings: - 100% success rate — All 28 models generated and applied AI textures without errors - Consistent quality — The futuristic Cobras design adapted well across different garment types - Color accuracy — Grey, black, and orange palette was maintained across all generations - Real-time rendering — Textures were applied and rendered in ~30 seconds per model - Bug fix validated — The race condition fix ensures textures persist correctly on all models